



**HKU
Med**

Upper Abdominal Pain: Peptic Ulcer, Gallstone, Pancreatitis

Prof. Wai Keung LEUNG

Division of Gastroenterology & Hepatology

Department of Medicine

School of Clinical Medicine

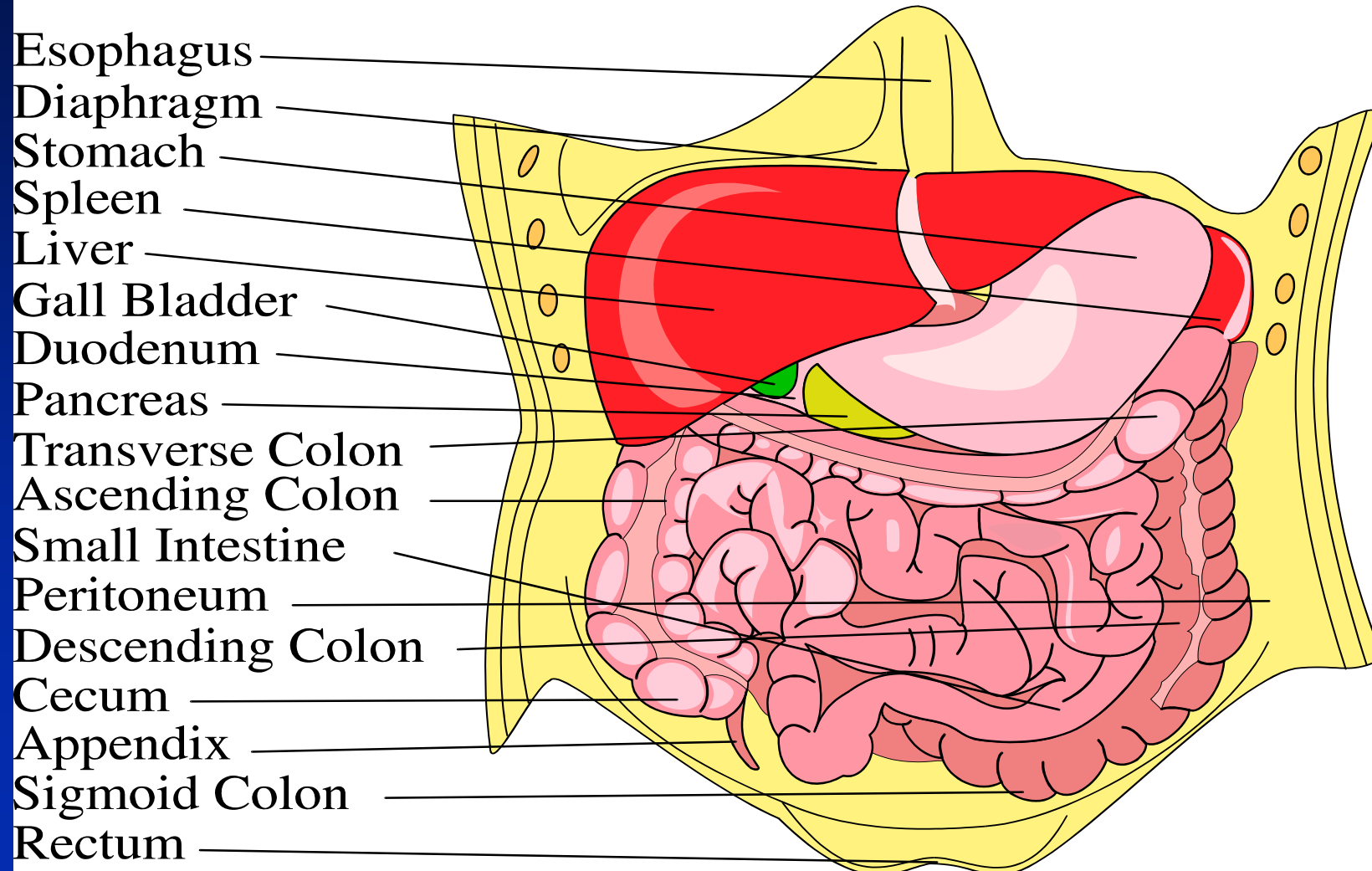
The University of Hong Kong

Outline

- Introduction
- Peptic ulcer
 - *Helicobacter pylori*
 - NSAID and aspirin
- Pancreatitis and gallstone

Abdominal Digestive Organs

Epigastric pain $\neq$ Peptic ulcer



Differential Diagnosis

- Gastroduodenal
 - Peptic ulcer (about 10%)
 - Atypical GERD
 - Gastric cancer (rare)
- Hepatobiliary
 - Pancreatitis
 - Ca pancreas
 - Gallstone
- Functional dyspepsia (about 60%)
- Drugs (NSAID, aspirin, aledronate etc)

Non-GI causes of epigastric pain

- Cardiac: MI, myocarditis etc
- Chest: pneumothorax, pulmonary embolism etc
- Hematological: acute leukemia, hemolytic anaemia...
- Metabolic: uremia, DM (DKA), porphyria, Addison's ...
- Toxins: hypersensitivity, lead poisoning..
- Infections: herpes zoster,
- Neurological: radiculitis, tabes dorsalis..
- Misc: muscular contusion, narcotic withdrawal, psychiatric disorder, heat stroke..

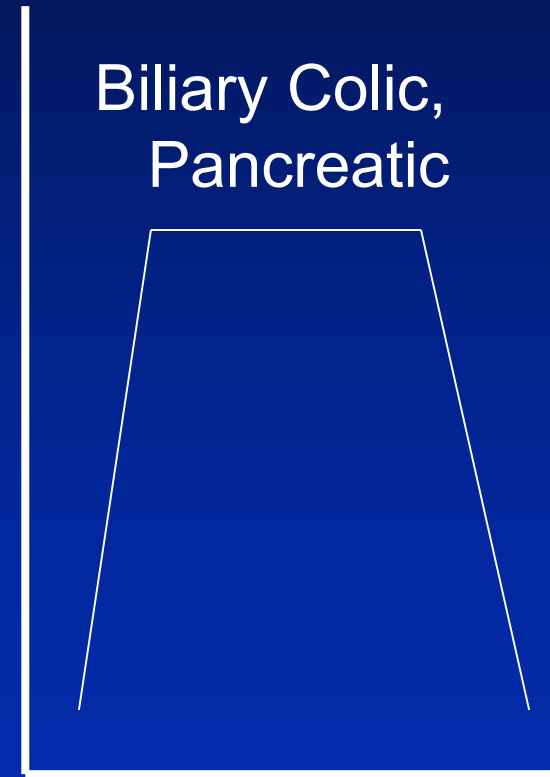
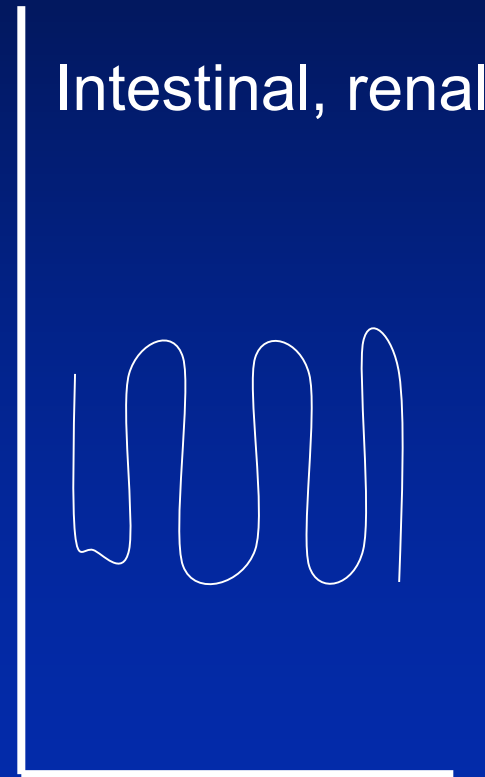
Approach to Upper Abdominal Pain

- History
- Physical examination
- Investigations

10 Questions on Pain

- **Location**
- **Referral**
- **Character**
(diffuse/localized)
- **Severity**
- **Duration**
- **Onset** (sudden/gradual)
- **Frequency**
- **Aggravating factors**
- **Relieving factors**
- **Associated symptoms**

Pain Severity



Acute epigastric pain referring to the back

- Posteriorly penetrating peptic ulcer
- Biliary pain
- Acute pancreatitis
- Dissecting aneurysm

Epigastric pain + repeated vomiting

- Food poisoning / Gastroenteritis
- Acute pancreatitis
- Gastric outlet obstruction

Agonizing pain but insignificant signs

- Acute pancreatitis
- Mesenteric thrombosis at early stage

Important considerations in history taking

- Duration of symptom
- Age
- Presence of alarming symptoms
- Drug history: NSAID/aspirin use
- Family history of CA stomach
- ?Alternative diagnosis (Non-GI causes)

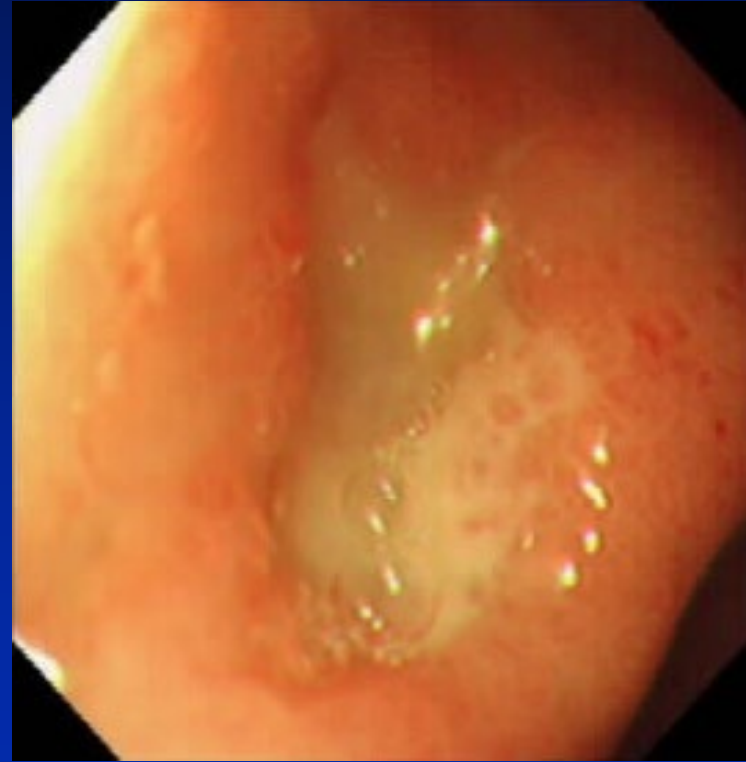
Alarming Features

- ≥ 45 years (new onset)
- **Anemia or bleeding**
- **Weight loss ($>10\%$ of BW)**
- Anorexia/early satiety
- Persistent vomiting
- Abdominal mass or lymphadenopathy
- **Dysphagia or odynophagia**
- Family history of upper GI cancer
- Previous gastric surgery or malignancy
- Previous history of ulcer
- Jaundice

Physical examination

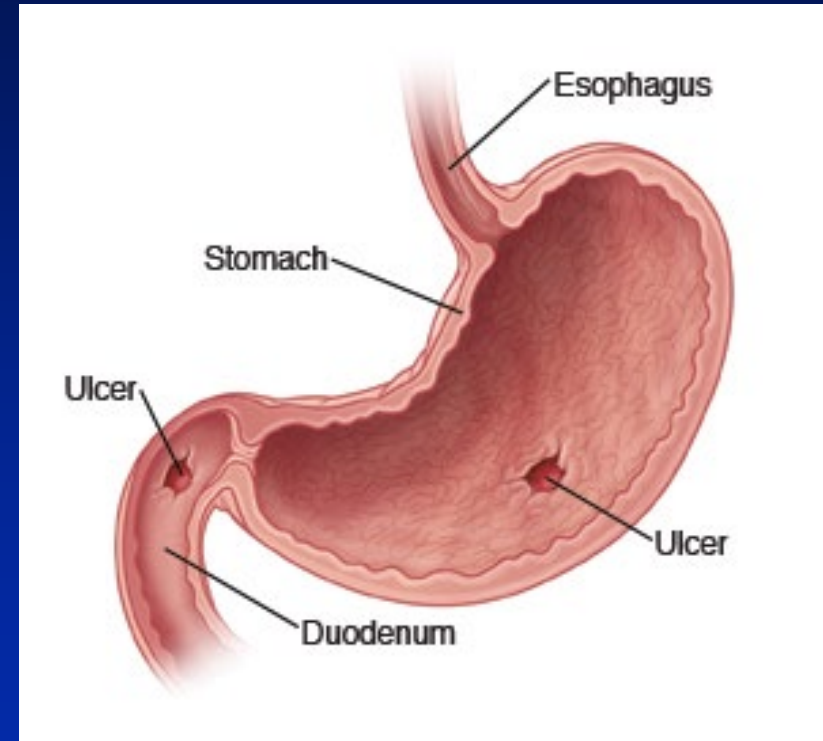
- General
 - Appearance, posture, in pain, vital signs
 - Fever, jaundice
- Abdominal
 - Location and degree of tenderness
 - Rebound tenderness, guarding, rigidity
- Genital, rectal and pelvic
 - PR exam, hernia orifices

Peptic Ulcer



Peptic Ulcer

- Gastric ulcer
- Duodenal ulcer
- Esophageal ulcer
- Meckel's diverticulum
- Anastomotic ulcer



Complications of ulcer

- Pain
- Bleeding
- Perforation
- Gastric outlet obstruction



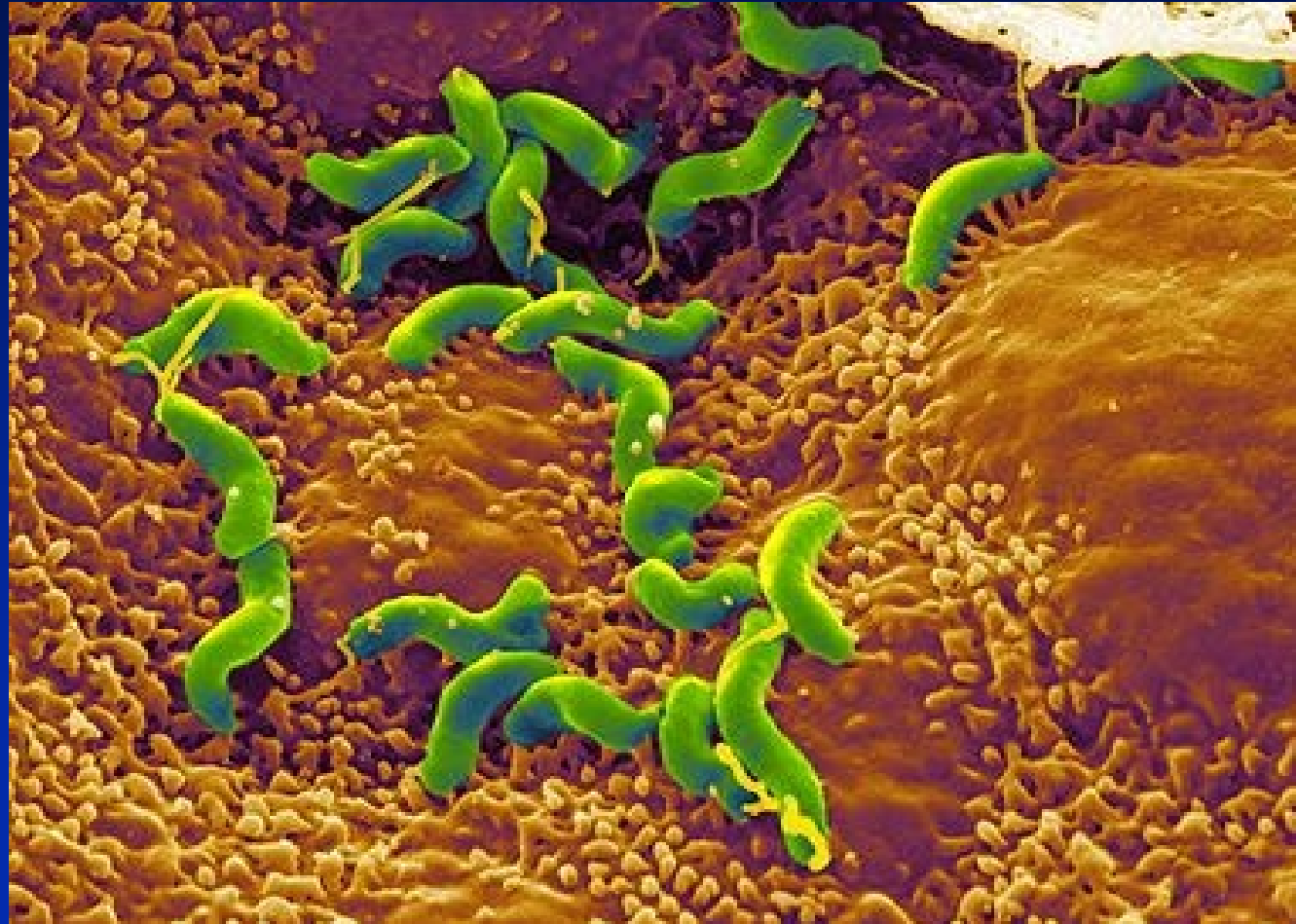
Causes of Peptic Ulcer

- *H. pylori* infection
- NSAID
- Aspirin
- Stress (ICU)
- Zollinger Ellison syndrome (Gastrinoma)
- Crohn's disease

Pharmacological Treatment of Peptic Ulcer Disease

- Acid suppression
- Eradication of *H. pylori*
- Avoid ulcerogenic drugs

H. pylori



- **Bacteria**

- Gram negative, rod, spiral shape with unipolar flagella.
 - Discovered in 1982 by Drs Barry Marshall and Robin Warren, Nobel laureate

- **Prevalence**

- Around 50% in adults in HK, decreasing worldwide
 - Developing countries 50-80%, Developed countries 10-30%
 - Increased with age, poor socioeconomic status, poor hygiene.

- **Transmission**

- Unknown, probably oral-oral route
 - water supply, poor sanitary condition, crowded environment
 - low socio-economic status

- **Colonization**

- Strong urease activity, hydrolyses urea to ammonia and bicarbonate, increase resistance to low pH of stomach.



The Nobel Prize in Physiology or Medicine 2005

"for their discovery of the bacterium *Helicobacter pylori* and its role in gastritis and peptic ulcer disease"



Photo: C. Northcott

Barry J. Marshall

① 1/2 of the prize

Australia

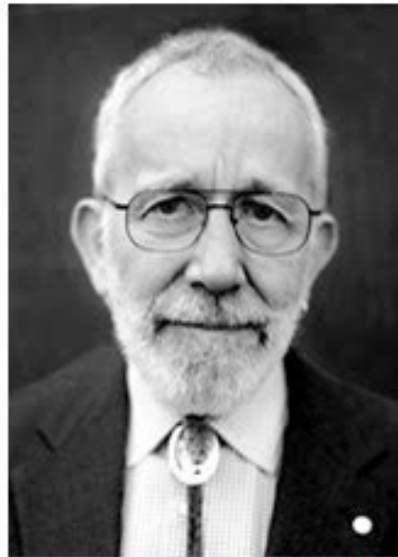


Photo: U. Montan

J. Robin Warren

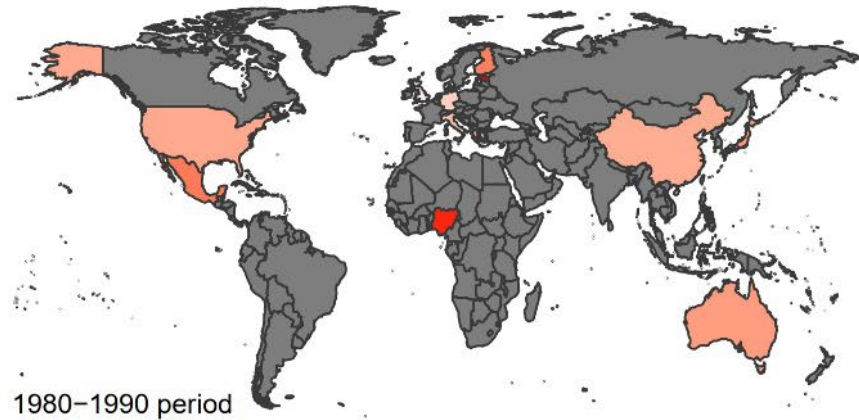
① 1/2 of the prize

Australia

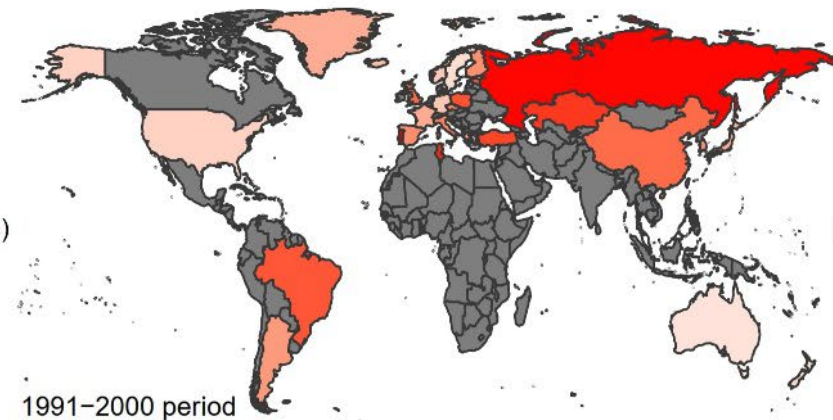
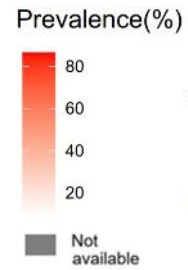


© REUTERS

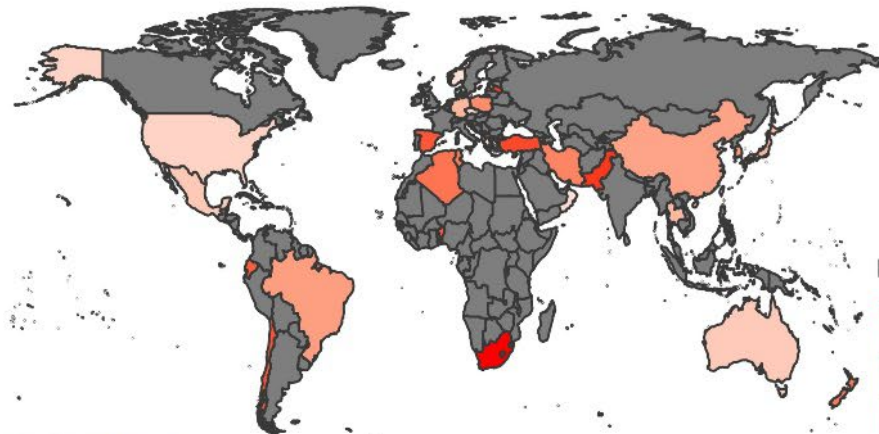
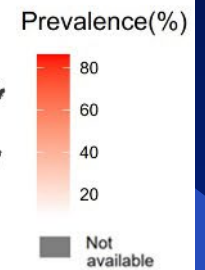
Updated HP Prevalence (1980-2022)



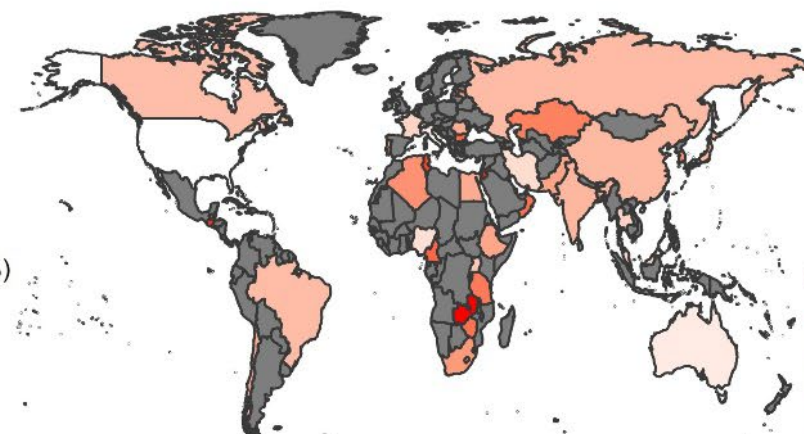
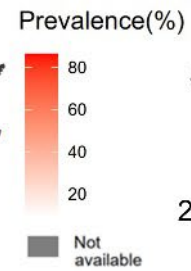
1980-1990 period



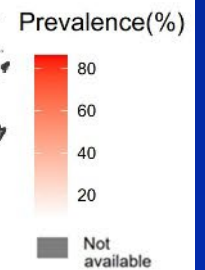
1991-2000 period



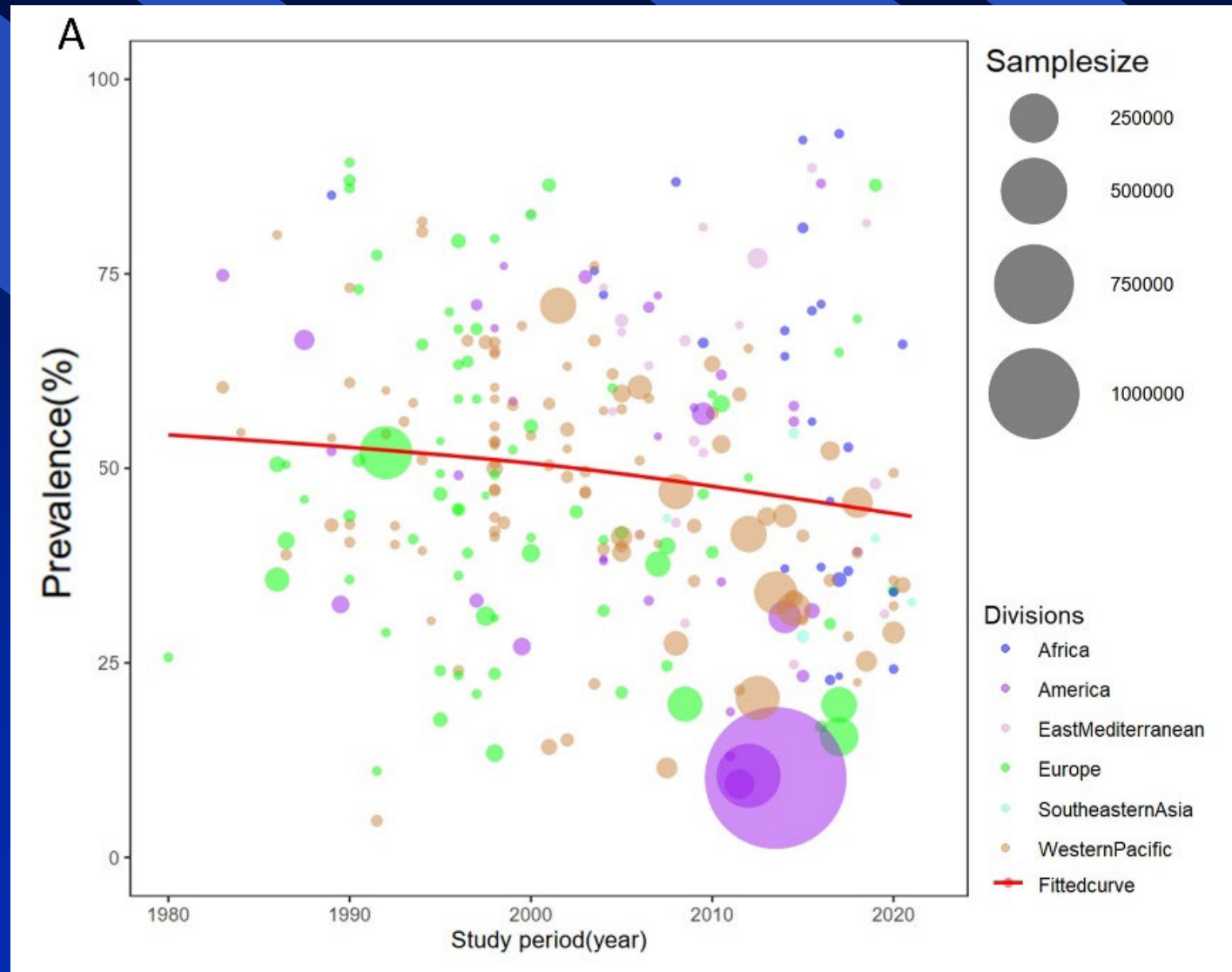
2001-2010 period



2011-2022 period



Temporal trend of *H. pylori* prevalence 1980-2022



H. pylori associated diseases

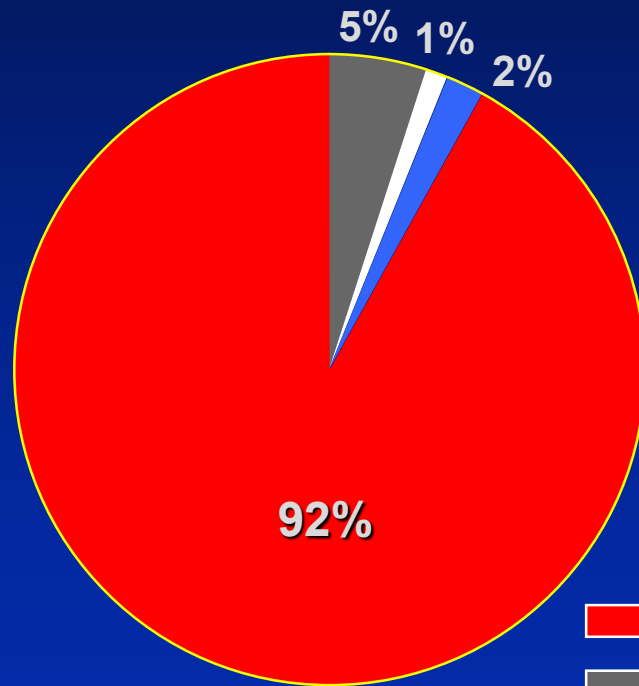
□ Gastroduodenal:

- Chronic gastritis
- Dyspepsia
- Duodenal ulcer
- Gastric ulcer
- Gastric cancer (adenocarcinoma)
- Gastric MALT lymphoma

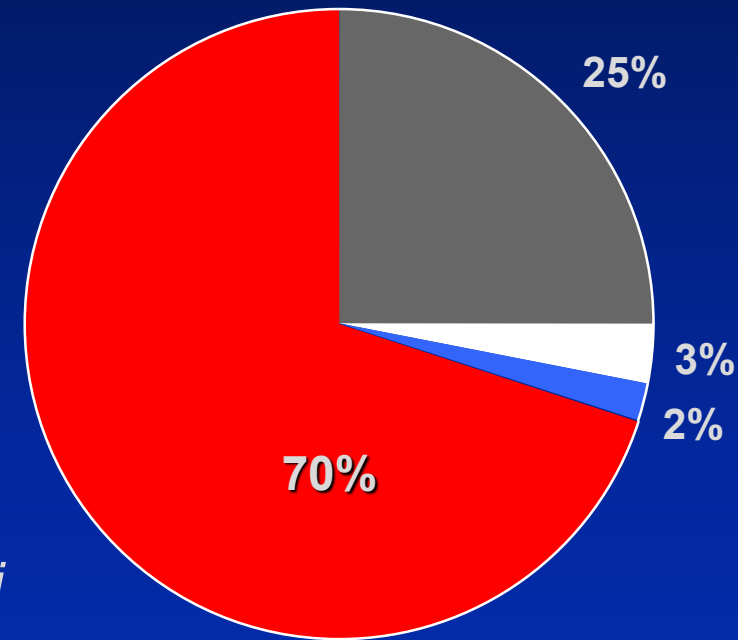
□ Extra-GI system: ? ITP, Anemia (B12 and Fe def)

H. pylori is a causal factor in most cases of peptic ulcer disease

Duodenal ulcer



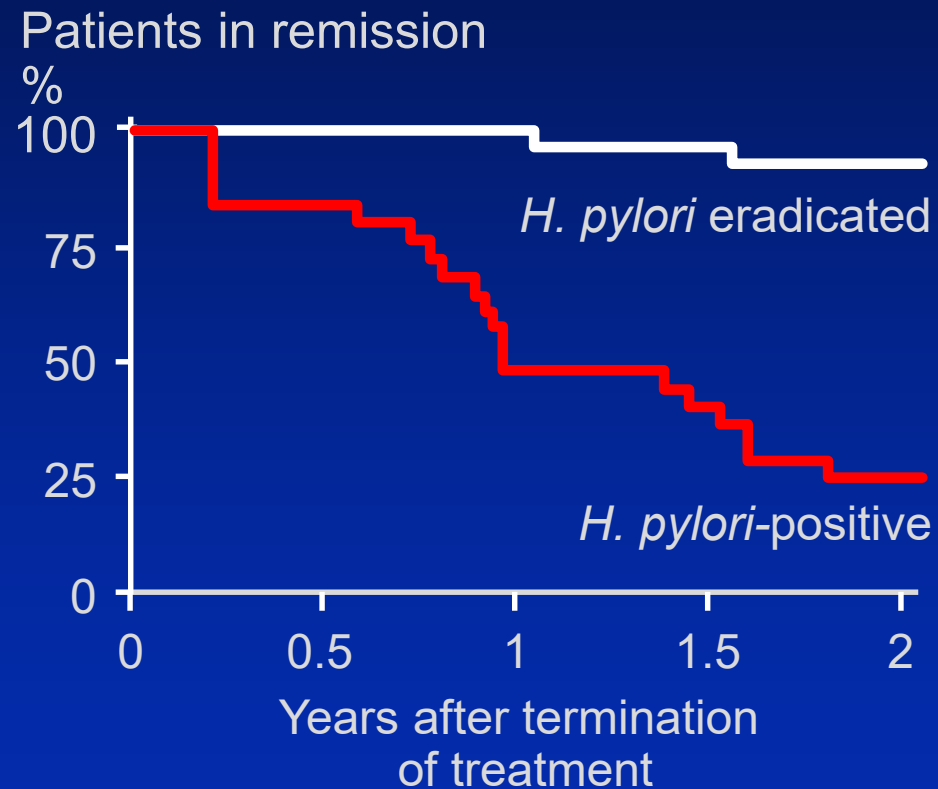
Gastric ulcer



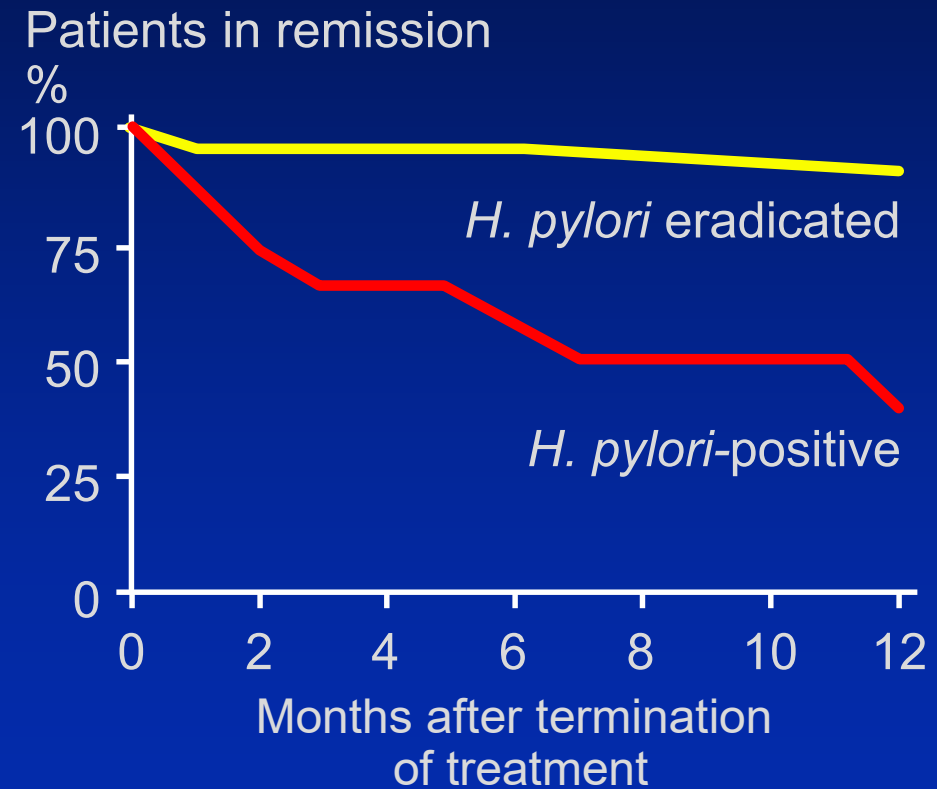
- *H. pylori*
- NSAID
- Cancer (Zollinger Ellison)
- Other

Eradication of *H. pylori* results in long-term remission of peptic ulcer disease

Duodenal ulcer



Gastric ulcer

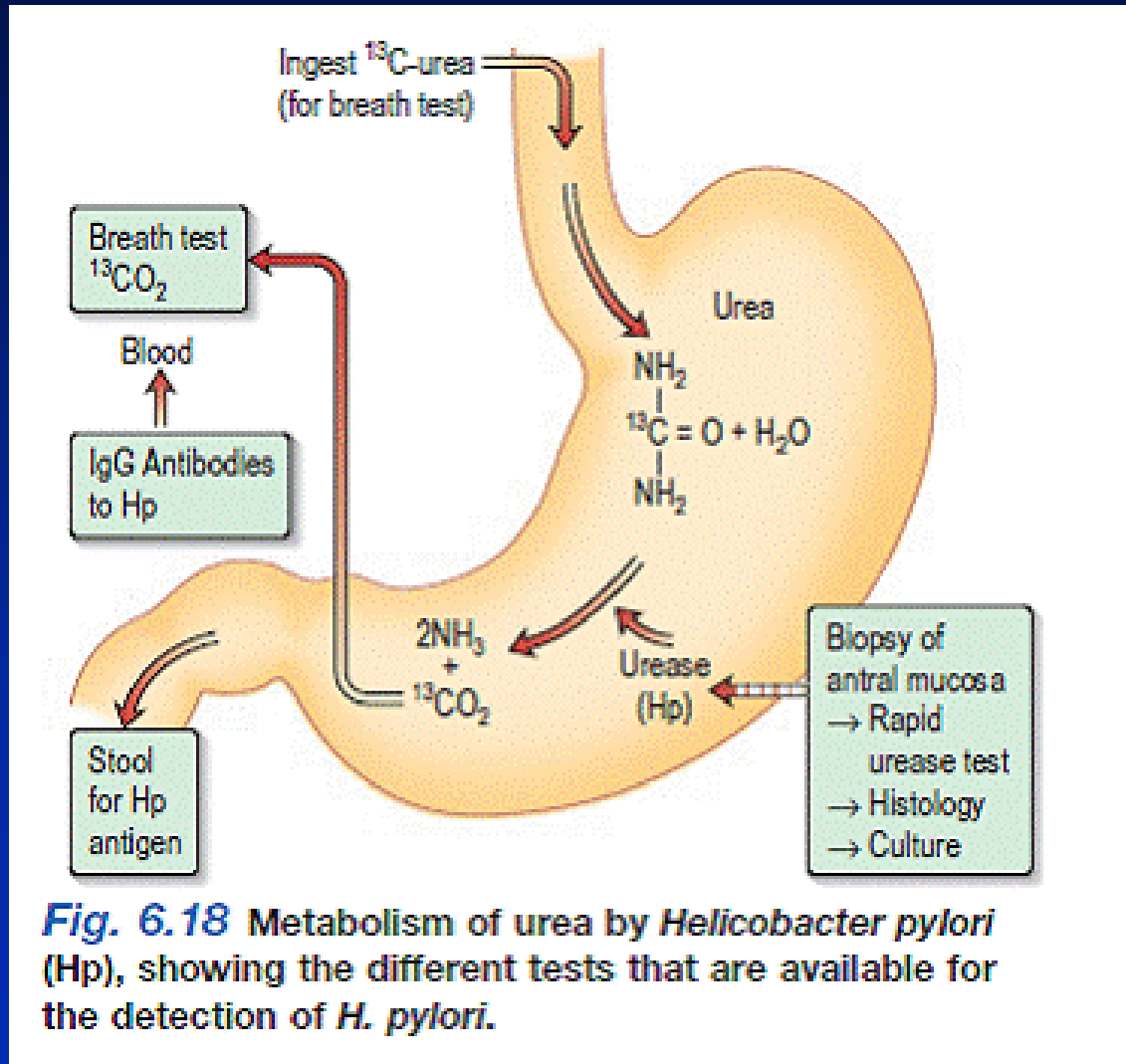


Diagnosis of *Helicobacter pylori*

- Invasive (require endoscopy and biopsy)
 - Rapid urease test or CLO test
 - Culture: antibiotic sensitivity testing
 - Histology: H&E. Special stains increase sensitivity
- Non-invasive
 - whole blood / serum test (antibody)
 - urea breath test (urease activity)*
 - Stool (antigen)*

*useful for monitoring after treatment

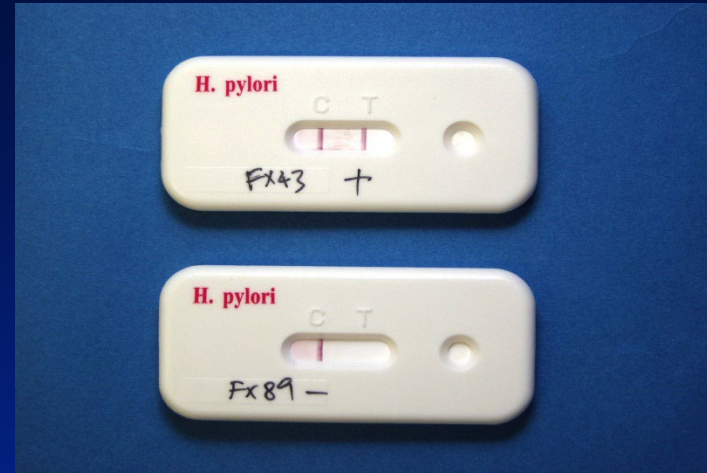
Urease Activity



Non-invasive HP testing



Urea Breath Test



Serology Test

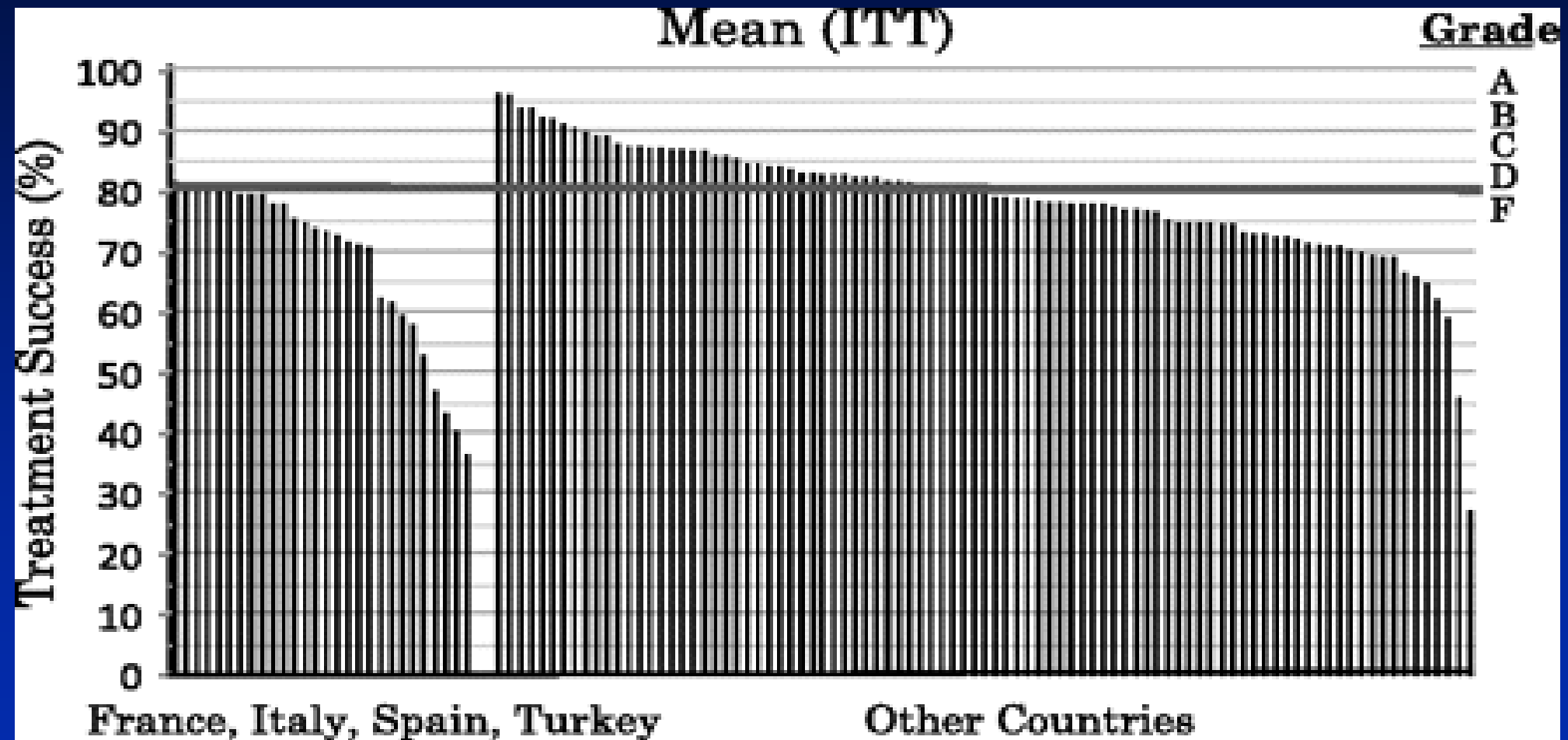


HP Stool antigen

HP eradication therapies:

- PPI-based triple therapy
 - PPI + Amoxillicin + Clarithromycin/Metronidazole
- Bismuth-containing quadruple therapy
 - PPI+ Bismuth + Tetracycline + Metronidazole
- Non-bismuth-containing quadruple therapy
 - PPI + Amoxillicin + Clarithromycin + Metronidazole

Falling efficacy of standard triple therapy



What governs eradication failure?



- Patient's factor
 - Allergic history
 - **Non-compliance**
 - Previous antibiotics use

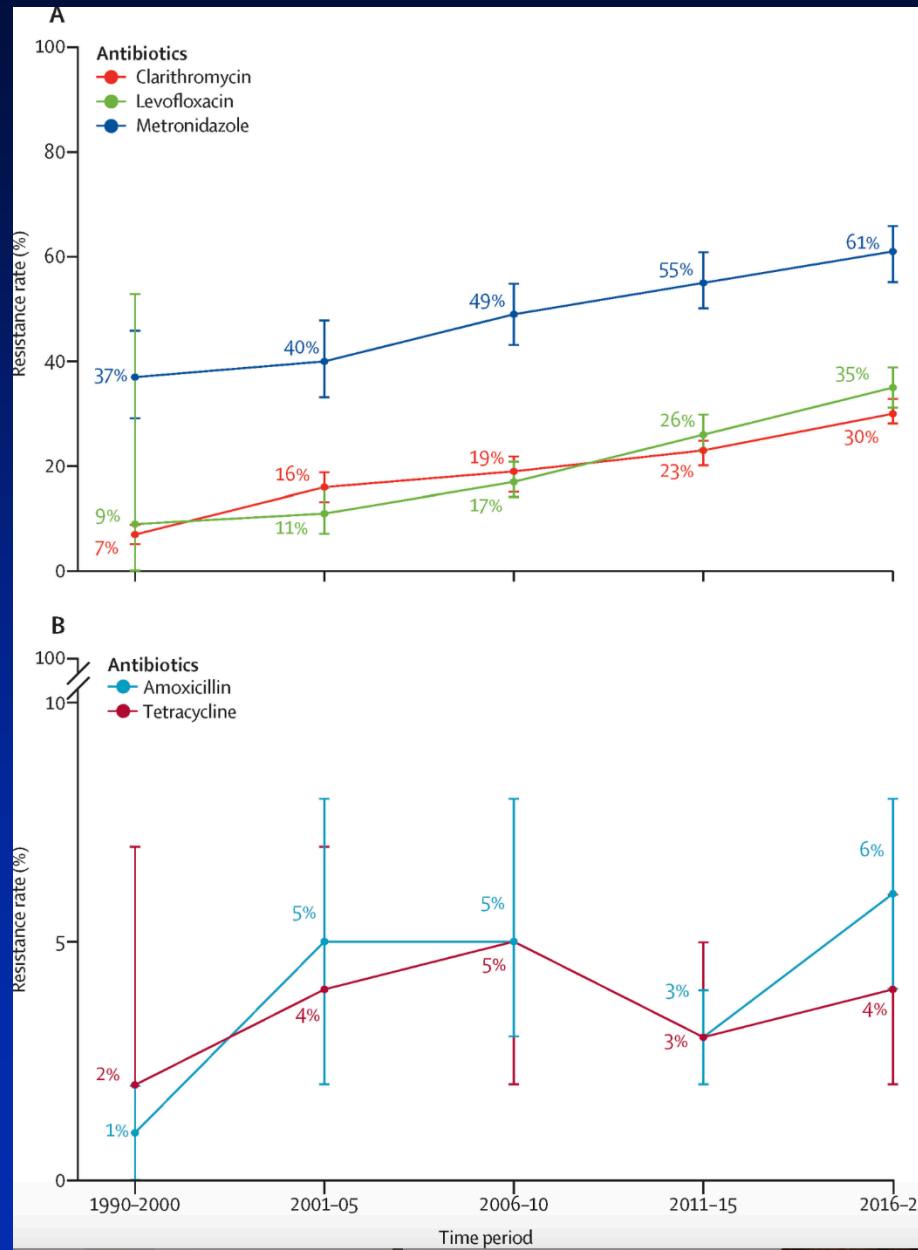


- **HP factor**
 - Antibiotic resistance

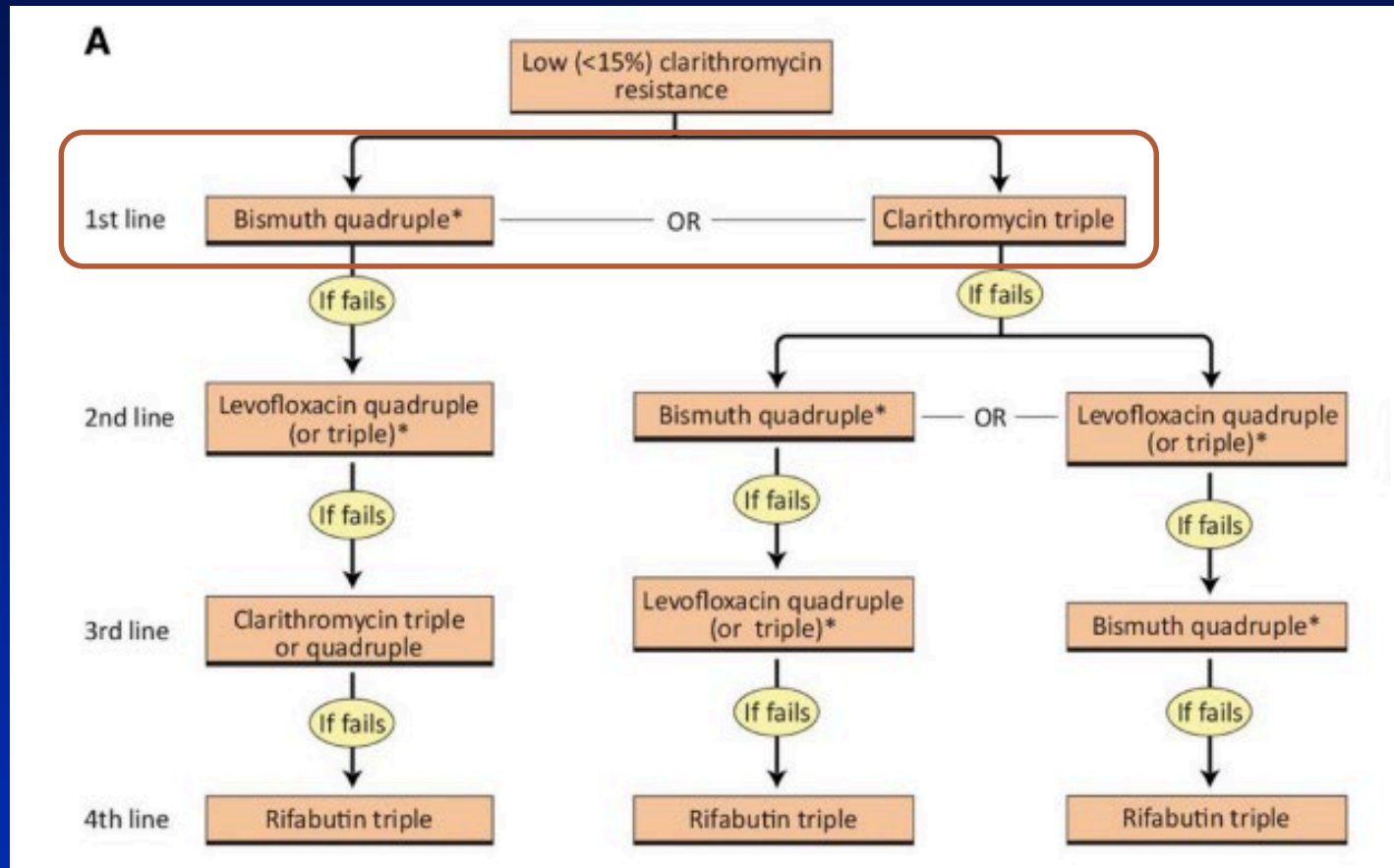


- Eradication regime factor
 - **Choice of antibiotics**
 - Degree of **acid suppression**
 - **Length** of eradication therapy

Prevalence of Antibiotic Resistance of HP in Asia Pacific Region



Maastricht VI Consensus (Low <15% CLA resistance)



BQT x 14 days, unless 10-day equally effective
Clarithromycin triple x 14 days

HK Consensus Treatment Algorithm: 1st line

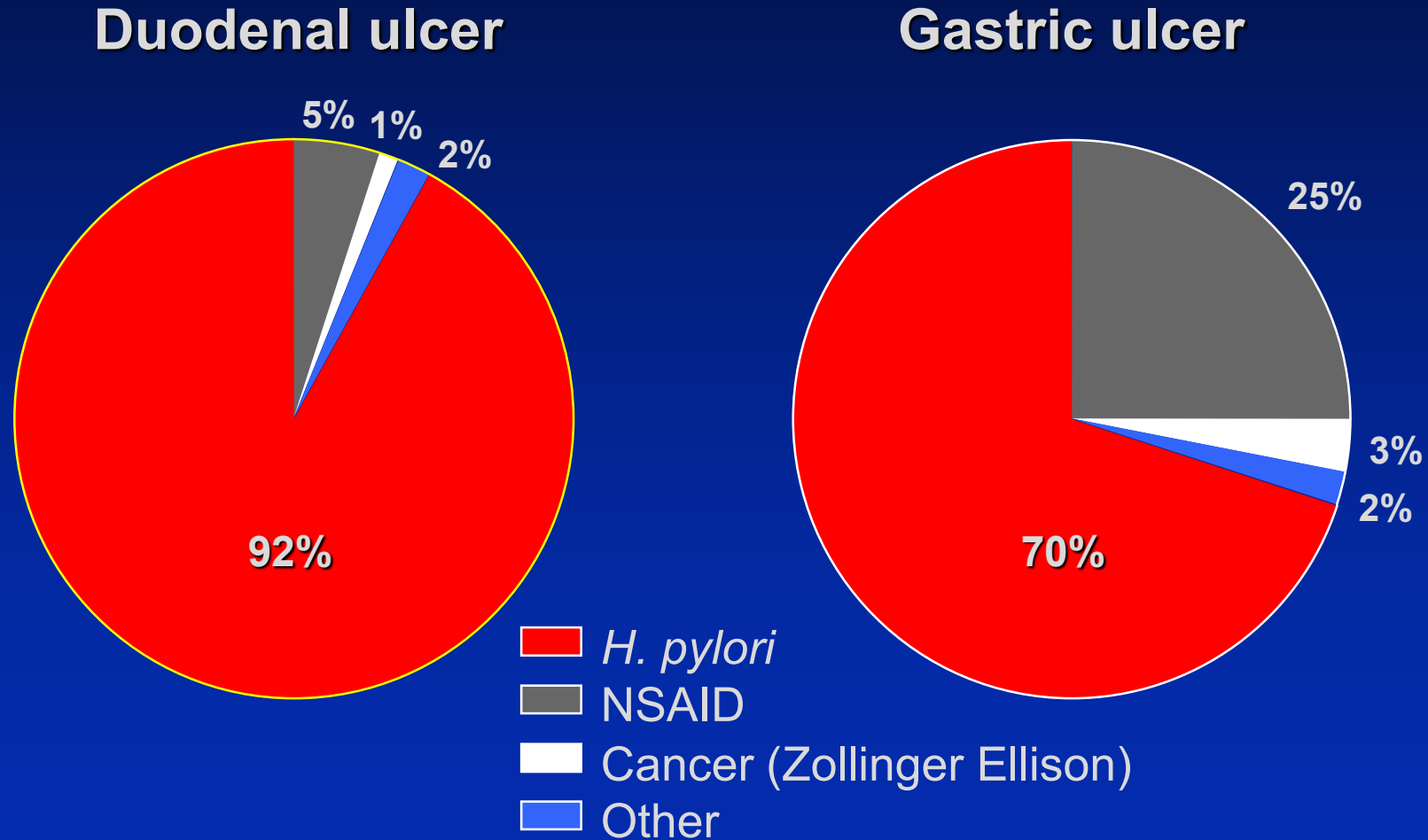
Statement 16: Therapies that can be considered in the first-line setting for *H. pylori* eradication include:

- a) **Triple therapy** with a proton pump inhibitor (PPI), clarithromycin, and amoxicillin for 14 days;
- b) **Bismuth quadruple therapy** with a PPI, tetracycline, metronidazole, and a bismuth salt for **10–14 days**.

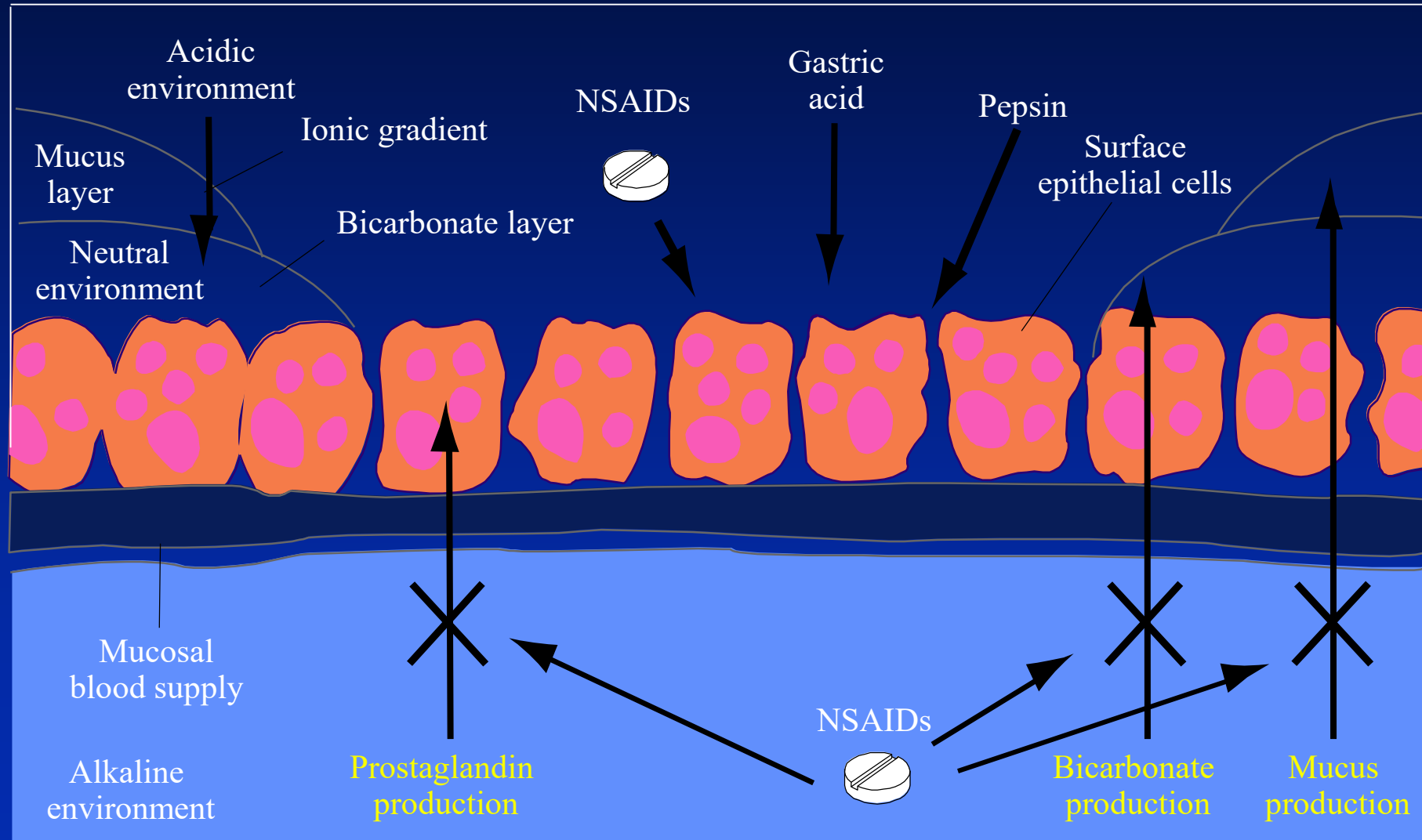
- Evidence quality: 1/2
- Grade of recommendation: strong/conditional
- Consensus level: 100%

NSAID and Aspirin

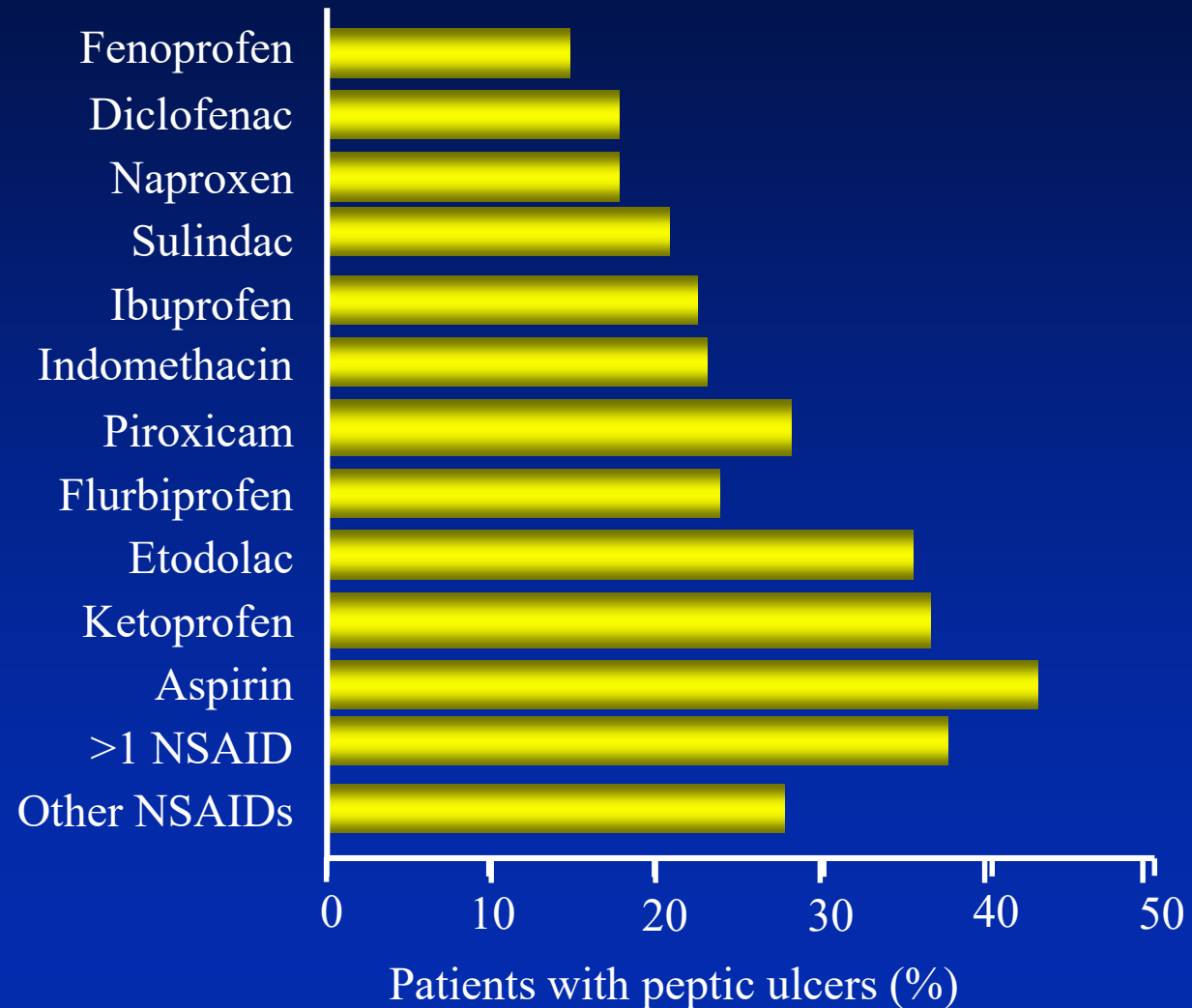
NSAID/aspirin is another important causal factor of peptic ulcer disease



NSAID & peptic ulcer



Prevalence of peptic ulceration is dependent on the relative NSAID toxicity





Risk factors for upper GI complications occurring with NSAIDs

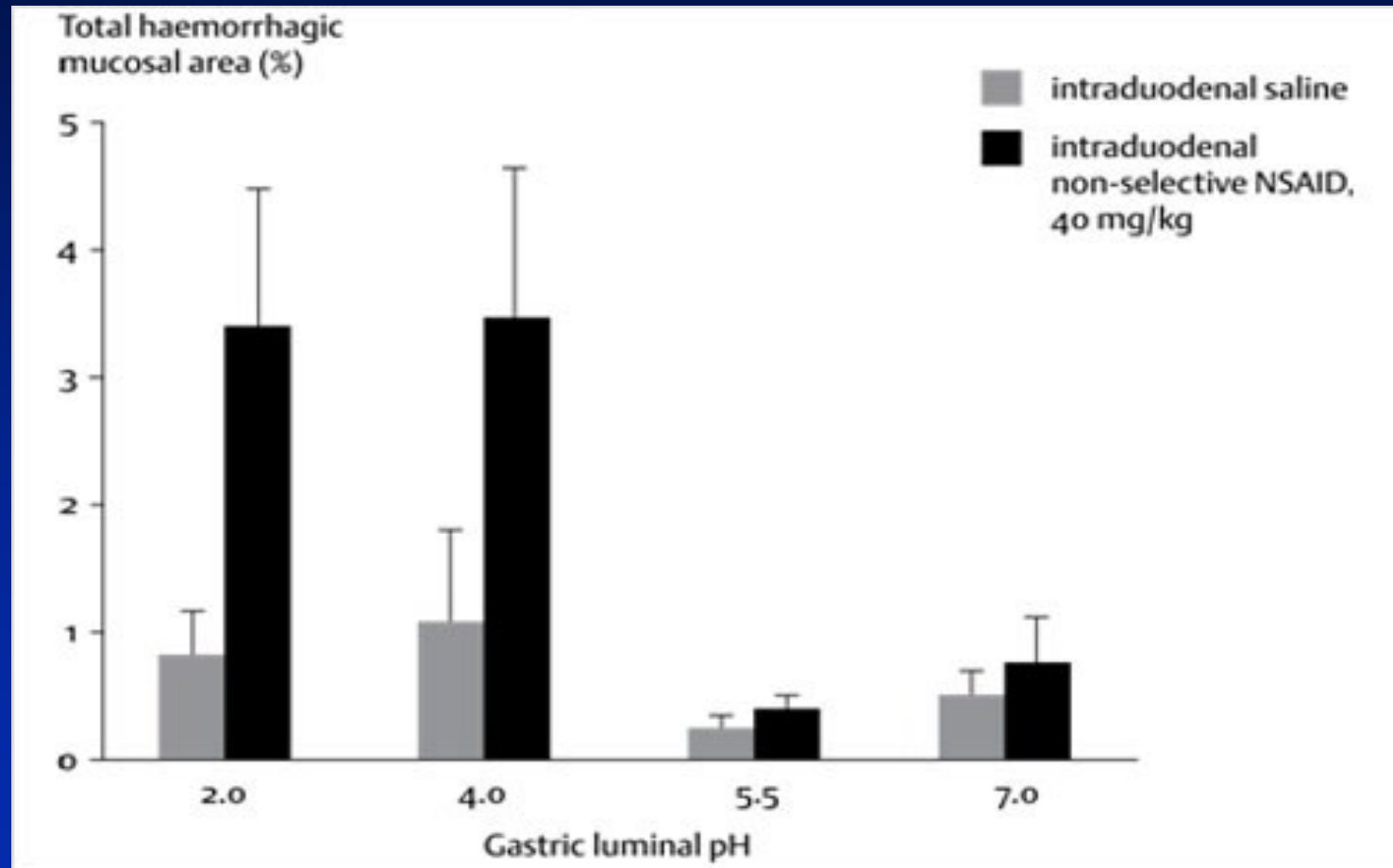
□ Patient-related factors:

- age >60 years
- history of peptic ulcer disease/upper GI complications.

□ Drug-related factors

- use of a relatively toxic NSAID
- use of a high dose of NSAID (or two NSAIDs used concurrently)
- concurrent use of an anticoagulant
- concurrent use of a corticosteroid.

Gastric mucosal damage by NSAID is pH dependent





Preventing NSAID-associated upper GI side-effects

- Review indications of NSAID
- Switch to COX-2 inhibitor
- HP eradication
- Co-therapy
 - PPI
 - H2RA
 - Misoprostol (PGE analogue)

How to prevent ASA related GI complications?

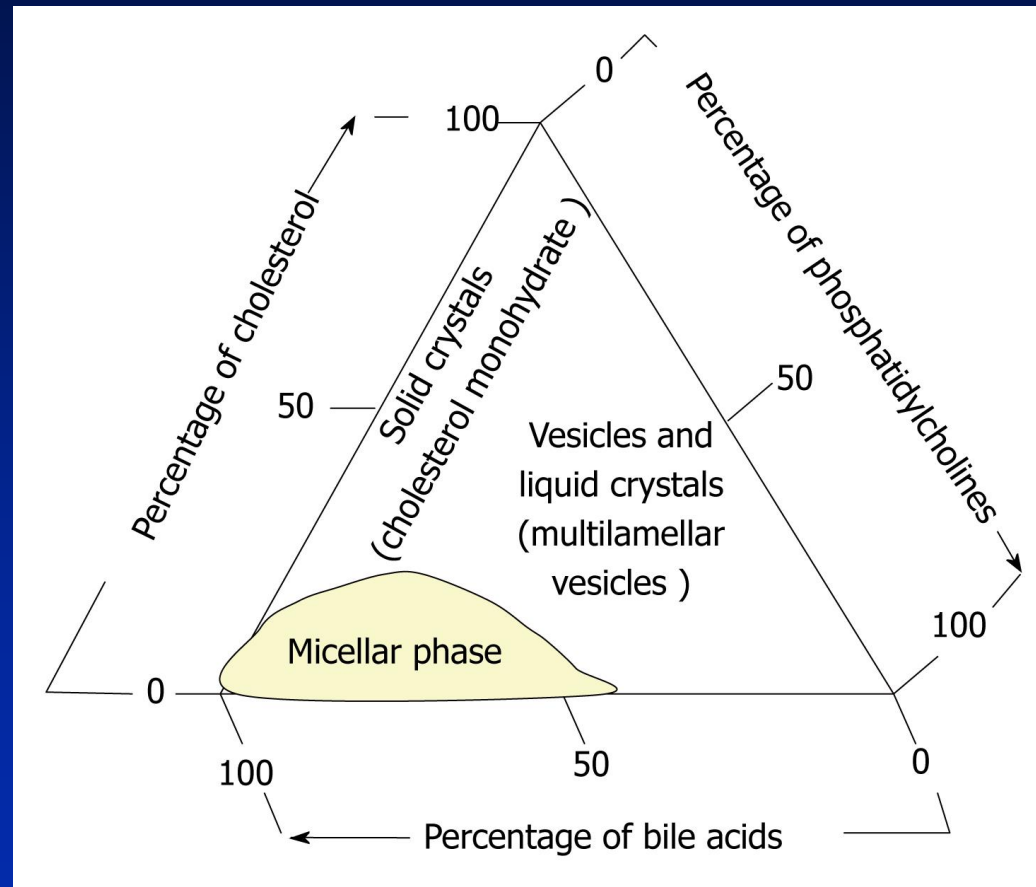
- *H. pylori* eradication
- H2 receptor antagonists
- Proton Pump Inhibitor
- Misoprostol
- Substitution with Clopidogrel?

Gallstones

Gallstones

- 10% of population above 40: Fat, Female, Forty
- 3 types- cholesterol, bile pigment, mixed
- Causes: high concentration of cholesterol/ bile salt, stasis, infection
- Risk factors:
 - Estrogen, OC pills, pregnancy, rapid weight loss in obese patient, use of fibrates, prolonged TPN, ileal resection or jejunioileal bypass, ileal disease (Crohn's disease), spinal cord injury, vagotomy, DM, thalassemia, cirrhosis

Pathogenesis of Gallstone



GS: Symptoms, Investigation and Tx

- Most asymptomatic
- Biliary colic, nausea, vomiting, dyspepsia.
- **Complications:** acute cholecystitis, gangrenous, empyema, perforation, cholangitis, pancreatitis, gallstone ileus
- Investigation: Ultrasound, MRI
- Treatment: asymptomatic gallstone- no treatment
- Acute cholecystitis - 90% settled with bed rest, antibiotics, elective cholecystectomy. Some require urgent surgery

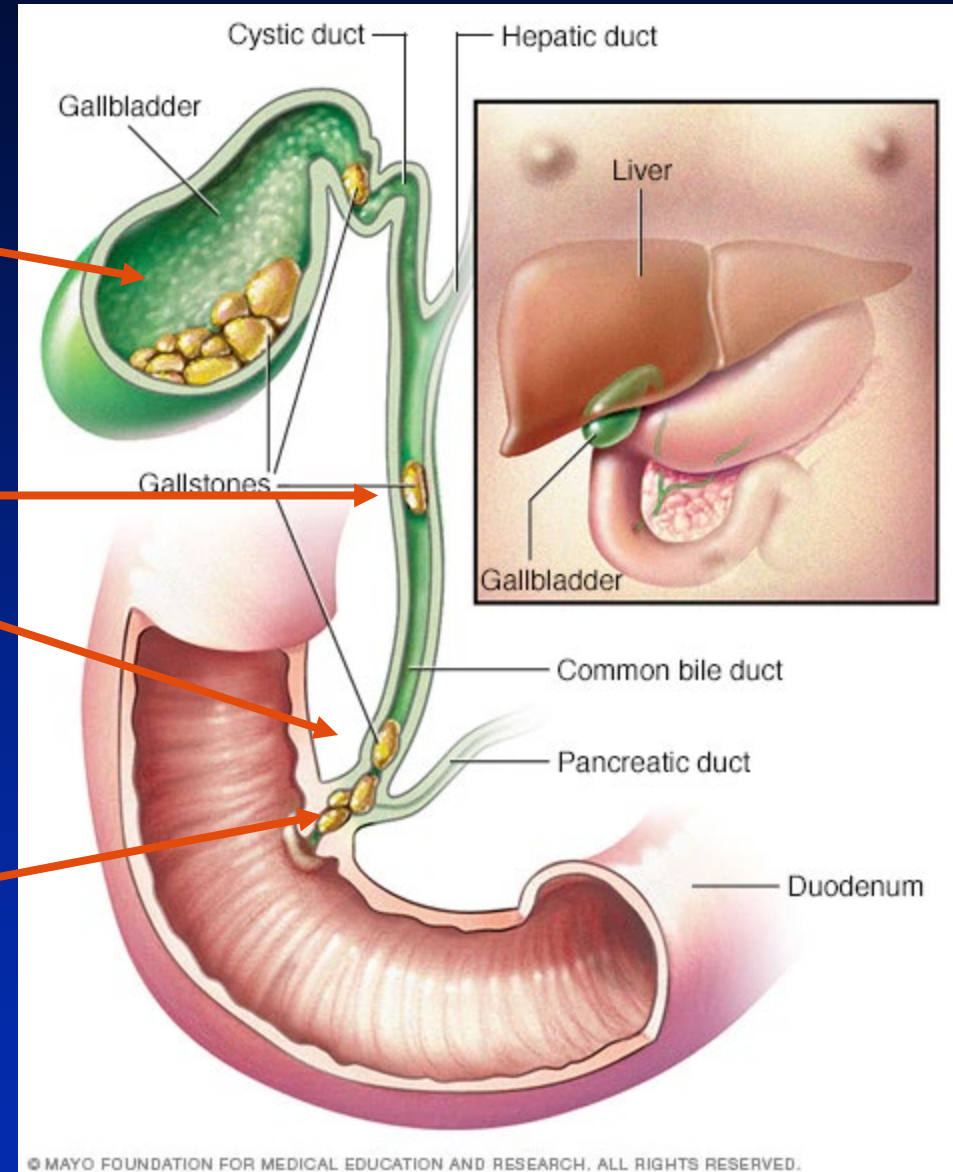
Ultrasound – Gallstone



Biliary colic
Cholecystitis

Cholangitis

Pancreatitis



Pancreatitis

Acute Pancreatitis: Aetiology

- Obstructive (biliary stones, others)
- Alcohol
- Infection: mumps, coxsackie, CMV, hepatitis A and B, mycoplasma, Legionella, Mycobacterium tuberculosis
- Drugs- steroids, azathioprine, 6-MP, thiazide, tetracycline, valproic acid, furosemide, sulfonamide, sulindac, acetaminophen, salicylates, erythromycin.
- Iatrogenic: post-surgery, **post-ERCP**,
- toxins, trauma, pancreatic tumours
- Hypertriglyceremia
- Idiopathic

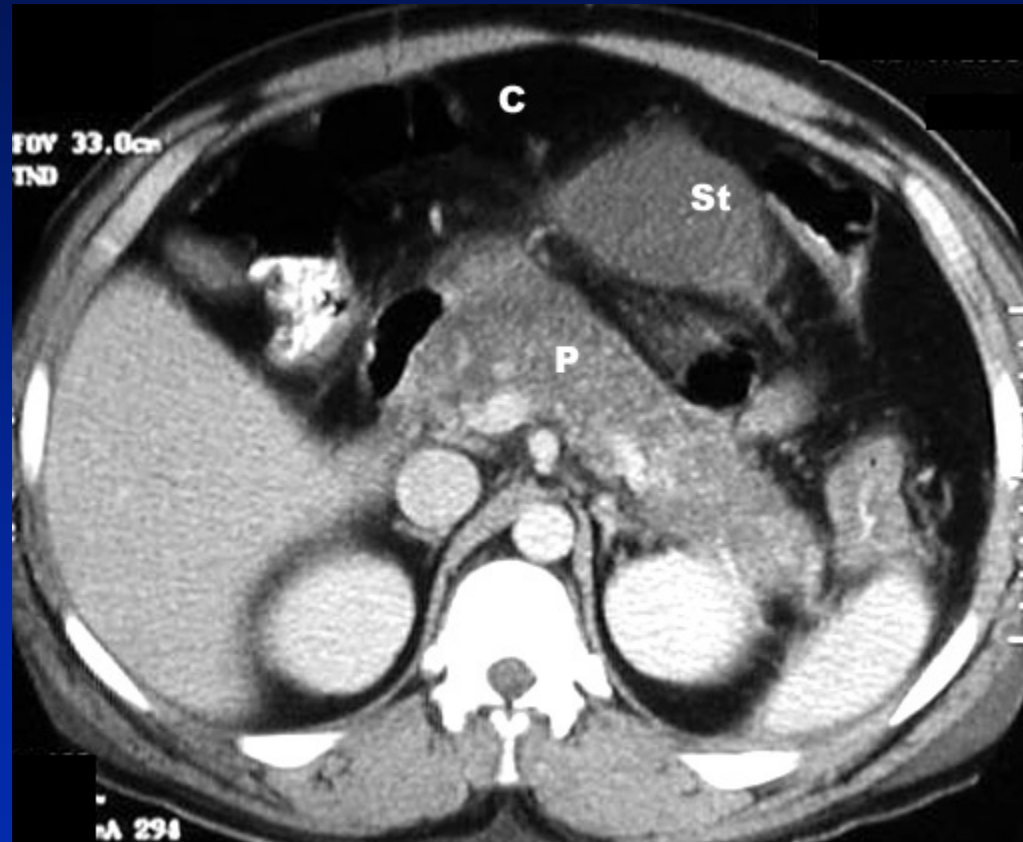
Diagnosis of acute pancreatitis

- Acute abdominal pain
- 3x increase in **serum amylase** and/or lipase
- Ultrasound for biliary pathology
- CT scan: assess severity, necrosis or complications like pseudocysts or abscess
- ERCP only for biliary and pancreatic duct pathology

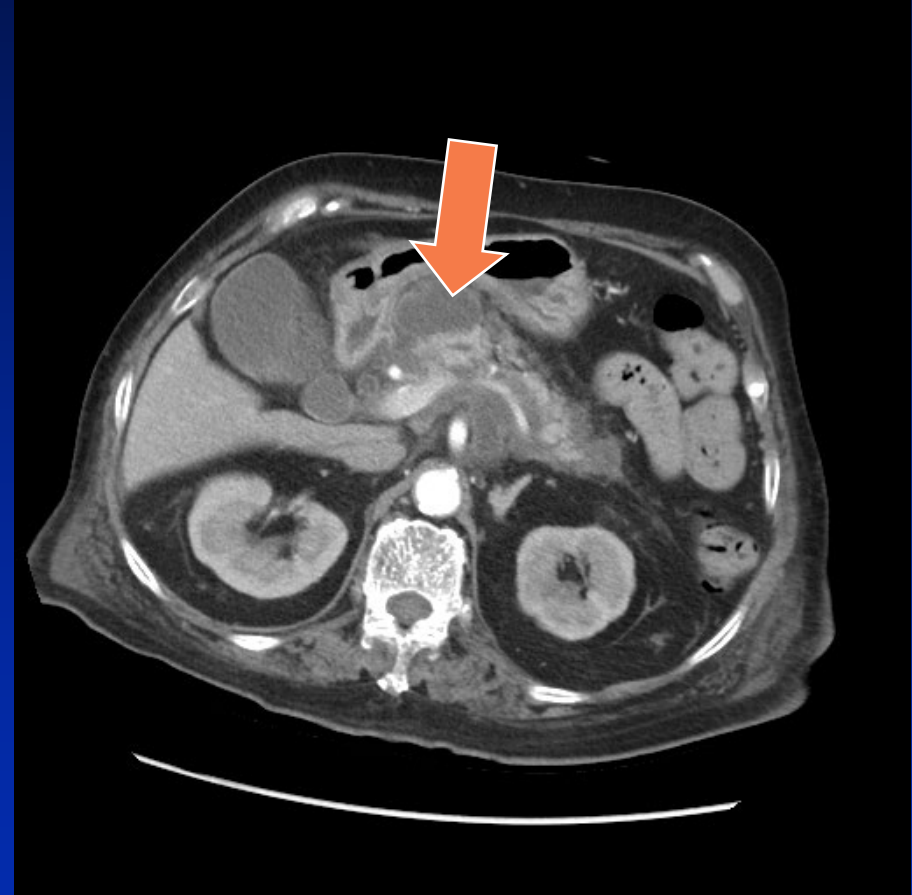
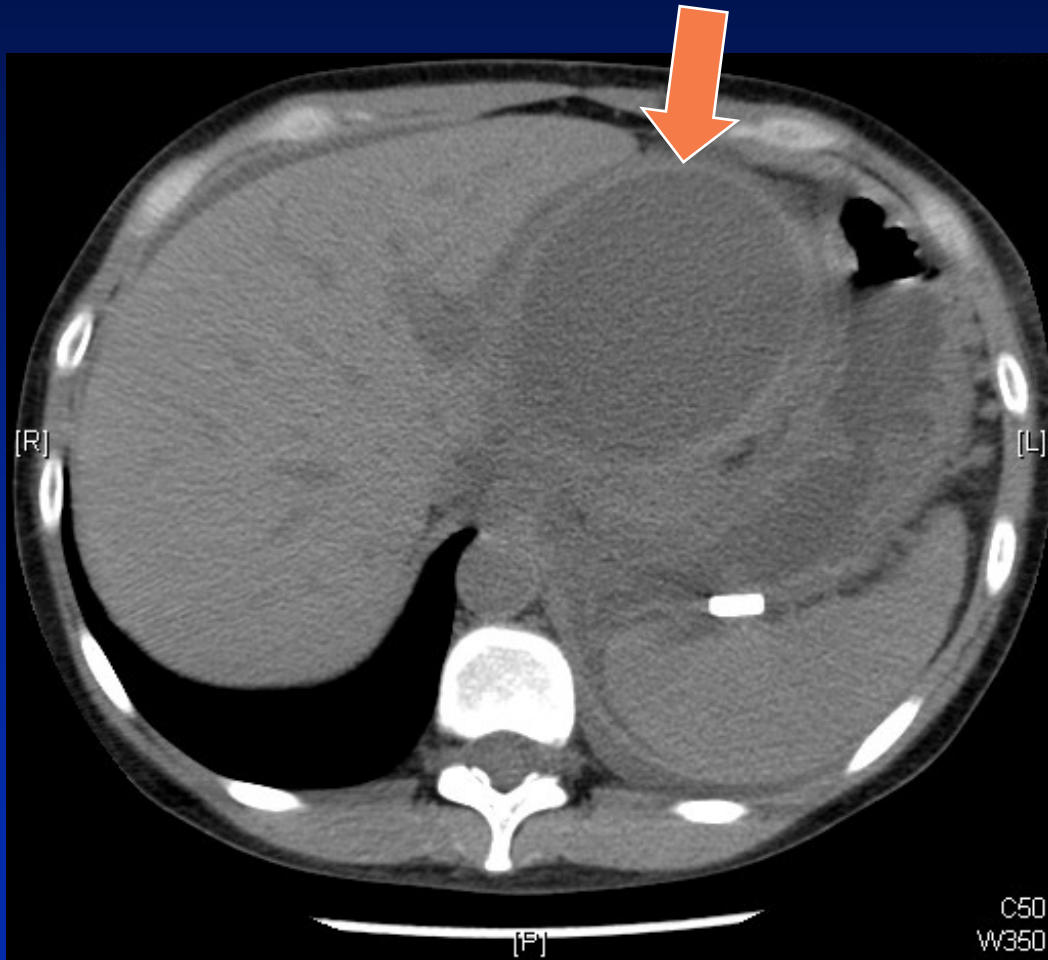
Prognostic scores

- Clinical: respiratory failure, peritonitis, shock, renal failure, GI bleeding indicate severe disease.
- Local complications: pseudocyst, abscess, necrosis.
- Revised Atlanta Classification
 - Mild (no organ failure, no local or systemic complications)
 - Moderately severe (transient organ failure <48h)
 - Severe (persistent organ failure >48h)

CT abdomen- pancreatitis



Pseudocyst



Treatment

- Mild: supportive, hydration, analgesics, NG suction for ileus or vomiting
- Severe: antibiotics, endoscopic or surgical debridement for infected necrotising pancreatitis
- ERCP: perform early for removal of biliary stones may decrease mortality
- Manage complications or organ failures

Chronic pancreatitis

- Etiology: **Alcoholic** (major cause), hypercalcemia, cystic fibrosis, trauma with residual duct injury, ampullary and duodenal diseases causing obstructing chronic pancreatitis, hereditary, automimmune, idiopathic.
- Classification:
 - Obstructive: fibrosis, ductal changes with dilatation.
 - Calcifying: fibrosis, intraductal stone, assoc with alcohol
 - Inflammatory: not the above two

Chronic pancreatitis

- symptoms: epigastric pain, weight loss, anorexia, steatorrhoea, DM, jaundice
- investigation:
 - AXR showed calcification, ultrasound showed pancreatic duct dilatation, parenchymal changes of increased echogenicity with irregular gland contour.
 - CT scan showed PD dilatation, calcification and cystic lesions.
 - ERCP showed ductal irregularity, dilatation, stenosis, stones
- treatment: stop drinking, pain control(maybe difficult), surgery in some,
- steatorrhoea- low-fat diet, pancreatic supplements eg pancreatin
- DM- diet control , drugs or insulin, difficult control because patients also lack glucagon.



Questions?

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.

(HKU Portal login required)